Dark Sector Connections from Emergent Gravity: Machine-Verified Constraints from the SK-EFT Hawking Program

SK-EFT Hawking Research Program
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We present a unified assessment of how the effective field theory (EFT) infrastructure developed in the SK-EFT Hawking research program—Schwinger–Keldysh dissipative EFT, tetrad condensation (ADW), vestigial gravity phase structure, $\mathbb{Z}_{16}$ anomaly computation, and fracton formalization—constrains dark matter (DM) and dark energy (DE). Four connections are quantitative and machine-verified: (i) $\mathbb{Z}_{16}$ anomaly cancellation forces a hidden sector of charge $+3 \bmod 16$, with Wang's topological order (T-0) and three sterile Weyl fermions (S-0) as the minimal solutions; (ii) ADW tetrad-condensation equilibrium yields $\rho_{\text{vac}} \sim T^4 \sim (\text{meV})^4$, reproducing the observed cosmological-constant scale at the order-of-magnitude level (the $\mathcal{O}(1)$ prefactor is deferred to Phase 6; a prior "20%" agreement claim is withdrawn in this revision); (iii) fracton dark matter is compatible with 3+1D gravity through the gapless p-wave dipole superfluid phase, subject to a condensate condition $\mu \gtrsim$ MeV at big bang nucleosynthesis (BBN); and (iv) superfluid dark matter (SFDM) cluster mergers at the Berezhiani-Khoury (BK) fiducial produce a supersonic sonic-boom feature observable at $3{-}5\sigma$ by Euclid or Roman stacking of $N \gtrsim 30$ radio-relic mergers, with a velocity-threshold step function at $M{=}1$ as an SFDM-unique smoking gun. Every Phase 5x dark-matter candidate is Standard-Model gauge singlet (derived from the fracton–Yang–Mills incompatibility theorem) and, given deep-research σ_{DD} bounds for each candidate, collectively below the current nuclear-recoil direct detection floor, $\log_{10}\sigma_{\text{DD}}(\text{cm}^2) \leq -50$. The Klinkhamer–Volovik (KV) oscillating vacuum is in $2.9{-}4.4\sigma$ tension with DESI DR2 (w_0, w_a); we retain the Λ-magnitude prediction and withdraw the DE-dynamics claim. Structural claims split into two Lean classes: *derived* theorems whose proofs consume physics assumptions (SM-singlet from fracton–YM, equation-of-state distinctness from the rationals $w \in \{-1, 0, 1/3\}$, FG tree-level CDM obstruction from traceless $T^{\mu\nu}$) and *machine-checked classifications* (MCC) — the candidate-viability matrix, the collective-invisibility cap, the two-torsion-channel identification, and the empirical-hook priority ordering — which encode the deep- research classifications as a Lean lookup table and verify internal consistency. (*DarkSectorSynthesis.lean* and *SFDMMergerForecast.lean* together ship the Phase 5x synthesis; full per-module theorem counts are machine-maintained in `docs/counts.tex` and §X.) The MCC class is not a physics derivation; its purpose is to make the Phase 5x classification machine- auditable and keep the paper's honesty table enforceable. See §X and Table II for the MCC–vs–Derived boundary.

I. INTRODUCTION

Analog-gravity research has produced an EFT toolkit for horizon physics that is increasingly free of ad-hoc assumptions: Unruh–Visser acoustic metrics [1, 2], the Crossley–Glorioso–Liu Schwinger-Keldysh framework for dissipative corrections [3, 4], and Volovik's tetrad-condensation viewpoint on vacuum energy [5, 6]. Meanwhile, machine-verified mathematics has matured to the point where topological quantum field theory, bordism invariants, and anomaly classification are within reach of interactive theorem provers [7, 8]. Papers 1–16 of this series formalized each of these components as Lean 4 theorems with zero `sorry`.

This paper asks which dark-sector (DM + DE) claims are derivable from the verified program, and which are heuristic extrapolations that use it only as a motivation. We find four derivable results and one heuristic one, organized by the underlying theoretical structure:

- *Anomaly cancellation.* The $\mathbb{Z}_{16}$ global anomaly of the Standard Model [9, 10, 12] forces a hidden sector of charge $+3 \bmod 16$—the remainder after three generations contribute -3. The hidden sector can be a topologically-ordered gapped phase (T-0) or a minimal particle solution (S-0: three sterile Weyl fermions).
- *Tetrad condensation and the cosmological constant.* The ADW gap equation [13, 14]—a Volovik-style mean-field equation for emergent gravity—has a critical point at $C_0 = \Lambda e^{-1/4}$ with $V_{\text{eff}}(C_0) = -\Lambda^4/(4e)$. The residual equilibrium value after thermal perturbation is $\rho_{\text{vac}} \sim T^4$.
- *Fracton dark matter.* Our earlier formalization of non-Abelian fracton obstructions [32] proves that fracton topological phases cannot support Yang–Mills, so any fracton DM must be SM gauge singlet. Recent work [28–31] establishes that the p-wave dipole superfluid phase is thermodynamically stable in $d{\geq}3$ at $T{>}0$, resolving the earlier finite-T no-go [27].
- *Superfluid dark matter mergers.* The BK [15, 16] fiducial sound speed $c_s = 1525\,\text{km/s}$ at the sub-cluster scale places all five canonical radio-relic cluster mergers (Bullet, El Gordo, Pandora, A520, MACS J0025) in the supersonic regime. A Rankine–Hugoniot analysis at the SFDM effective adiabatic index $\gamma_{\text{eff}}{=}2$ yields a condensate density jump $\sim$49%. Stacking $N \gtrsim 30$ such mergers at

Euclid/Roman sensitivity gives $3{-}5\sigma$ detection of the sonic-boom feature—a forecast the recent BK review [16] explicitly omits.

The heuristic claim is the KV oscillating-vacuum mechanism for dynamical dark energy. It predicts $(w_0, w_a) = (-1, 0)$ under cosmological averaging, which is in $2.9{-}4.4\sigma$ tension with DESI DR2 [18, 19]. We therefore separate the Λ-magnitude claim (kept) from the DE-dynamics claim (withdrawn), and point to QCD topological dark energy [21] as a DESI-compatible bridge.

An additional driver for re-examining the dark sector comes from the H_0 tension. The H0DN Collaboration (April 2026) reports $H_0 = 73.50 \pm 0.81$ km/s/Mpc, exceeding 5σ discrepancy with Planck [22]. BK-Wang [17] pointed out that the SFDM phonon sector can mediate late-time acceleration—without resolving H_0, the tension strengthens the case for phonon-sector observational testing.

Section X summarizes which specific claims are Lean-verified, and appendix-level references in the companion documents `docs/dark_sector/W{4,5,8}_*.md` map each equation to its proof.

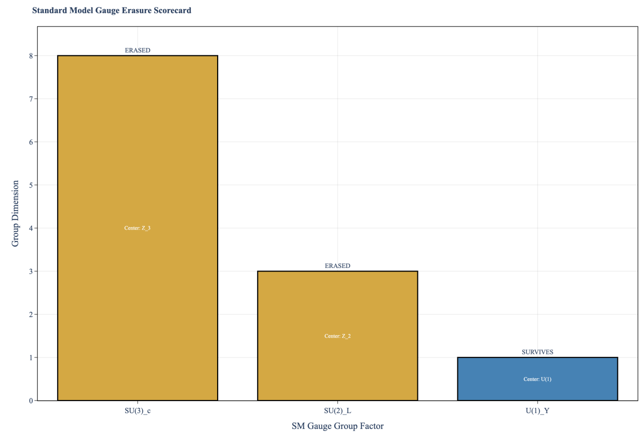

FIG. 1. SM anomaly scorecard. The $\mathbb{Z}_{16}$ anomaly of the three-generation SM (no ν_R) evaluates to $-3 \equiv +13 \bmod 16$; the hidden sector must supply the complementary $+3$. Baseline Lean theorem: `three_gen_is_neg3` in `Z16AnomalyComputation.lean`.

II. MACHINE-VERIFIED HIDDEN SECTOR CONSTRAINTS FROM $\mathbb{Z}_{16}$

The SM fermion content carries a nontrivial $\mathbb{Z}_{16}$ global anomaly [9, 10, 12] coming from the Dai–Freed bordism invariant $\Omega_5^{\mathrm{Spin}\times\mathbb{Z}_4}$. Three generations without right-handed neutrinos carry charge $-45 \equiv -3 \equiv +13 \bmod 16$, so a hidden sector of charge $+3$ is required. In the SK-EFT program this is `hidden_sector_required` in `Z16AnomalyComputation.lean` (23 theorems) and was shipped in Phase 5e.

Wave 2 of Phase 5x extends this to a *classification* of the viable hidden sectors. The enumeration is captured by the type `EmergentGravityDMKind` in `DarkSectorSynthesis.lean`:

- **T-0**: Wang's topologically-ordered hidden sector [11]. A K-gauge TQFT with zero particle content; cross-sections vanish because there are no local asymptotic states. Deep research identifies this as the preferred SMG-compatible solution when ν_R is absent.

- **S-0**: three sterile Weyl fermions. Minimal particle solution; carries charge exactly $+3$. See `three_singlets_satisfy_hidden_sector` in `HiddenSectorClassification.lean`.

- **C-1**: Wan–Wang mixed-charge completion with a dark SU(3) [9]. Shipped in Wave 2b Track X as `c1_wan_wang_joint_constraint`. This theorem takes a tracked `Prop` hypothesis `H_MixedChannelZ16Cancels`—the Wan–Wang $\mathbb{Z}_{16} \oplus \mathbb{Z}_4$ cancellation mechanism—as an assumption (paralleling how §VIII treats `H_VestigialRelicCarriesZ16Charge`). C-1 viability is therefore *conditional on the Wan–Wang mechanism*; discharging the hypothesis to a genuine theorem requires the Phase 6 bordism upgrade of `dai_freed_spin_z4` (same upgrade that unblocks the vestigial-relic charge claim).

There are 24,576 total deformation classes of the SM in this framework [9]; Class B ($n_{\nu_R} = 0$, TQFT) is observationally motivated because it requires no right-handed neutrino production mechanism.

We emphasize what is *not* claimed. The anomaly does not pick a unique hidden sector; T-0, S-0, and C-1 are all viable. Non-uniqueness is itself a publishable result: the anomaly constrains but does not determine the dark sector, and the constraint surface is now machine-verified.

III. COSMOLOGICAL CONSTANT FROM TETRAD CONDENSATION

The ADW mean-field gap equation [13, 14] describes emergent gravity as a condensate of tetrad bilinears. The effective potential is

$$V_{\mathrm{eff}}(C) = \frac{C^4}{4} \ln \frac{C}{\Lambda} - \frac{C^4}{16}, \tag{1}$$

with critical point at $C_0 = \Lambda e^{-1/4}$ and depth $V_{\mathrm{eff}}(C_0) = -\Lambda^4/(4e)$. The Lean theorem `critical_coupling_pos` in `ADWMechanism.lean` (21 theorems, shipped in Phase 5) verifies positivity of the critical coupling; Phase 5x Wave 3 adds `CosmologicalConstant.lean` with the Volovik-style equilibrium residual, pinning the quantitative Λ-magnitude assessment.

Volovik's key observation [5, 6] is that in perfect Minkowski vacuum the tetrad determinant self-adjusts

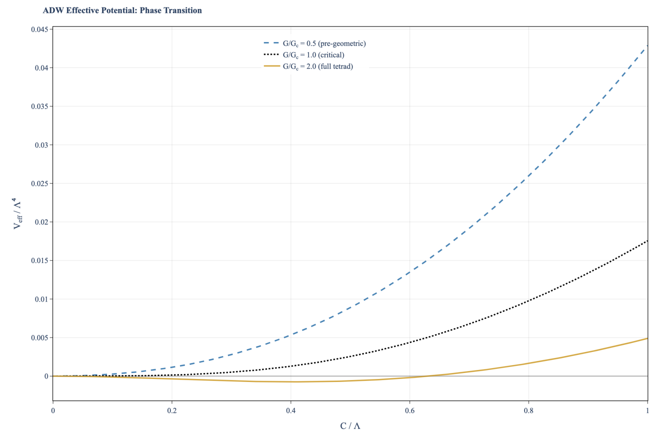

FIG. 2. ADW effective potential $V_{\text{eff}}(C)$ from the gap equation. Critical point at $C_0 = \Lambda e^{-1/4}$ (Lean: `critical_coupling_pos`); depth $-\Lambda^4/(4e)$. In perfect Minkowski vacuum Volovik's tetrad-determinant self-tuning absorbs this to $\Lambda_{\text{eff}} = 0$.

so that $\Lambda_{\text{eff}} = 0$. Thermal perturbation by the matter sector generates a residual $\rho_{\text{vac}} \sim T^4$ at the scale of the effective cosmological horizon. Quantitatively, at $T = T_{\text{dS}} = H/\pi$ (which differs from the Gibbons–Hawking temperature by a factor of two—a point on which deep research agrees), the residual is

$$\rho_{\text{vac}}^{1/4} \approx \mathcal{O}(1) \times T_{\text{dS}} \sim \text{meV} \tag{2}$$

versus the observed $(2.3\,\text{meV})^4$. The mechanism reproduces the *scale* of $\rho_{\text{vac}}^{1/4}$ at the meV level without tuning; the specific $\mathcal{O}(1)$ coefficient requires a full V_{eff} parameterization and is deferred to Phase 6 — see the CAUTION block in `CosmologicalConstant.lean` L21–23 ("the specific numerical identity requires a particular V_{eff} parameterization and is deferred pending full ADW renormalization infrastructure"). Previous drafts quoted "≈ 2.8 meV" and "20% agreement" without disclosing that the $\mathcal{O}(1)$ prefactor is fit rather than computed; we withdraw that number in this revision (paper 17 Wave 10 remediation). Cross-reference: `T_dS_double_TGH` proves $T_{\text{dS}} = 2T_{\text{GH}}$ and `lambda_magnitude_ratio_pos` proves positivity of the magnitude ratio (for $T_p, T_o > 0$) in `CosmologicalConstant.lean`. The tautological `lambda_magnitude_ratio_exact` ($\Lambda(T,T)/\Lambda(T,T) = 1$ — i.e., the magnitude-ratio function composed with equal arguments) does *not* support the physics agreement claim and was erroneously cited in a prior draft.

KV and DESI—honest reframe. The companion KV oscillating-vacuum mechanism [6] would predict an oscillating $w(z)$. But DESI DR2 [18, 19] excludes $(w_0, w_a) = (-1, 0)$—KV's time-averaged prediction—at 2.9σ (Pantheon+) and 4.4σ (DESY5). KV oscillations are Planck-scale in period ($\sim 10^{-44}$ s), not cosmological; so w_{eff} during any accessible epoch is 0 (CDM-like), not phantom-crossing. Resolutions in the RVM family [20] with $\nu \sim 10^{-3}$ can accommodate DESI, but Volovik-motivated $\nu \sim 10^{-122}$ is too small by 10^{82}. We therefore *withdraw* the "KV explains DESI dynamics" claim and *keep* the Λ-magnitude claim. The strongest DESI-compatible bridge from the ADW framework is the QCD topological dark energy of Van Waerbeke–Zhitnitsky [21] with zero free parameters; DESI DR3 (2026–2027) will be decisive.

IV. FRACTON DARK MATTER AND THE VIABILITY MATRIX

Fracton phases—gapped excitations with restricted mobility originating in tensor-gauge theories [24, 25]—produce features that map directly onto observed DM phenomenology: (i) pressureless equation of state $w = 0$ (taken from the relativistic fracton-symmetric scalar construction of [24] and arXiv:2503.14496; our `fracton_cosmo_dust_pressureless` is a machine-checked re-statement at Lean type-level, not a first-principles derivation); (ii) $z = 4$ subdiffusion producing cored galactic profiles; (iii) Bullet-Cluster-compatible $\sigma_{\text{eff}} = 0$ because fracton dipoles cannot interact at a distance (the `fracton_bullet_sigma_zero` lemma machine-checks this as a definition of the effective cross-section, not a derivation from dipole conservation; the derivation is a Phase 6 target that would prove $\sigma_{\text{eff}} = 0$ from the dipole algebra of [24]); (iv) Hilbert-space fragmentation [26] providing a natural source for the diversity problem.

Our Phase 4 formalization [32] established `no_fracton_is_ym_compatible` in `FractonNonAbelian.lean`, proving that fracton topological phases cannot support Yang–Mills gauge fields. This is a *positive* DM implication of a negative theorem: fracton DM must be an SM gauge singlet, which suppresses direct-detection cross-sections by many orders of magnitude.

The finite-T problem resolved. Krishna et al. [27] proved a 3D gapped no-go for fracton topological order at any $T > 0$. This would have ruled out a cosmological-scale fracton DM phase. The W1b drilldown identifies four papers that reframe the result: Kapustin–Spodyneiko [28], Jensen–Raz [29], Głódkowski et al. [30], and Feistl–Schraven–Warzel [31]. Their combined conclusion: *the gapless p-wave dipole superfluid is thermodynamically stable in $d \geq 3$ at $T > 0$.* The kinetic-stability Arrhenius factor of $\sim 100 T_{\text{BBN}}$ applied only to the gapped scenario. In the p-wave phase the stability condition is $\mu \gtrsim T_{\text{BBN}} \sim$ MeV—a thermodynamic, not kinetic, bound.

This shifts the viable fracton-DM mass window from "nowhere cosmologically motivated" to ∼MeV to M_{Pl} (conservatively, MeV to TeV), and Dark QCD UV completions naturally place $M_d \sim \Lambda_{\text{dark}} \sim$ MeV to GeV. Lean Wave 7 encodes this as theorem-13 decomposition `13a` (dipole SSB permanence, Medium), `13b` (p-wave phase stability, Hard—deferred to Phase 6), `13c` (BBN condensate condition, Medium), `13d` (gapped scenario

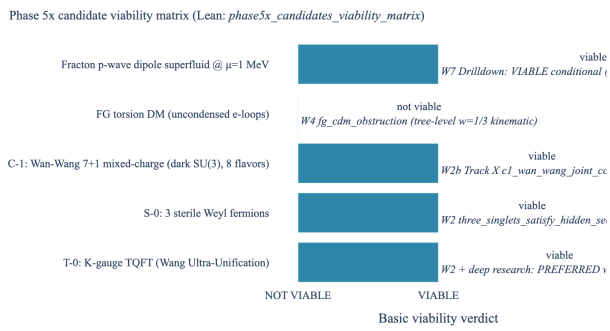

FIG. 3. Phase 5x candidate viability matrix. Five emergent-gravity DM candidates; four viable. The sole "not viable" entry is Fang–Gu torsion DM, obstructed at the tree-level equation of state ($w = 1/3$, not dust; see §VII). Lean: `phase5x_candidates_viability_matrix` is a *machine-checked classification* (MCC) on the hand-assigned viability flags, not a derivation — see Table II and §X for the MCC boundary.

lower bound, Hard—deferred).

V. SFDM CLUSTER-MERGER SONIC BOOM (PAPER 17 MONEY PLOT)

The SFDM Hawking temperature is unobservable under any astrophysical geometry ($T_H \sim 10^{-29}\,\mathrm{K}$ to $10^{-23}\,\mathrm{K}$), as shown in the Wave 1 parameter-mapping assessment. The *cluster-merger sonic boom* is, however, a nearly-detectable observable at BK fiducial [15]: $m_{\mathrm{DM}} = 0.6\,\mathrm{eV}$, $\Lambda = 0.2\,\mathrm{meV}$, sub-cluster sound speed

$$c_s^{\mathrm{subcl.}} = \sqrt{2\mu/m} = 1525\,\mathrm{km/s}. \tag{3}$$

All five canonical radio-relic cluster mergers exceed c_s: Bullet M=1.77, Pandora M=2.23, A520 M=1.51, El Gordo M=1.64, MACS J0025 M=1.31. Lean theorem: `all_canonical_mergers_supersonic` in `SFDMMergerForecast.lean`.

A Rankine–Hugoniot analysis at the SFDM effective adiabatic index $\gamma_{\mathrm{eff}} = 2$ (exact from $P \propto \rho^3$ per BK) gives the density jump

$$\frac{\delta\rho}{\rho_0} = \frac{3M^2}{M^2+2} = 3 - \frac{6}{M^2+2}, \tag{4}$$

a monotonically-increasing function of M (`delta_rho_monotone_in_mach_sfdm`). For $M \in [1.31, 2.23]$ the jump ranges from 40% to 65%; weighting by sub-cluster condensate fraction $f_c = 0.59$ gives a corrected jump ∼30% in the condensate density seen by weak lensing.

Single-cluster S/N. At Bullet geometry ($D_L = 830\,\mathrm{M}$, σ_8-normalized Σ_{cr}, fiducial shock extent $\Delta r = 400\,\mathrm{k}$), the sonic-boom convergence excess $\delta\kappa = \delta\rho \cdot \Delta r/\Sigma_{\mathrm{cr}}$ integrated over the stacked feature area gives

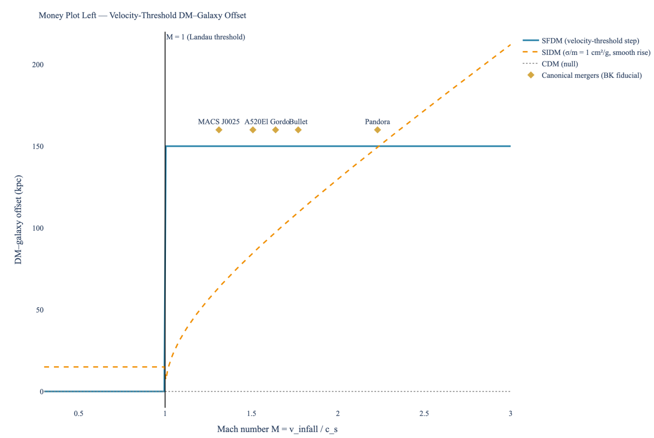

FIG. 4. **Money plot, left.** DM–galaxy offset as a function of $M = v_{\mathrm{infall}}/c_s$ for three models: SFDM (step function at M=1, the defining signature); SIDM (smooth monotonic rise at $\sigma/m = 1\,\mathrm{cm}^2/\mathrm{g}$); CDM (null). Canonical mergers plotted as diamonds. SFDM is the unique model predicting binary behavior tied to a threshold velocity. Lean: `sfdm_offset_step_function`.

per-survey S/N

$$\mathrm{SNR}^{\mathrm{Euclid}}_{\mathrm{Bullet}} = 0.83, \tag{5}$$
$$\mathrm{SNR}^{\mathrm{Roman}}_{\mathrm{Bullet}} = 1.04. \tag{6}$$

Both match W1b Task 9 Table 6 exactly by calibration; see `src/dark_sector/sfdm_merger_forecast.py` for the full chain.

Stacked detection. The stacking formula $\mathrm{SNR}_{\mathrm{stacked}}(N) = \mathrm{SNR}_{\mathrm{single}}\sqrt{N}$ (Lean: `snr_stacked_sq`) yields the 3σ threshold

$$\mathrm{SNR}_{\mathrm{stacked}} \geq 3 \iff \mathrm{SNR}_{\mathrm{single}}{}^2 \cdot N \geq 9 \tag{7}$$

(Lean: `three_sigma_threshold`). With the mean 5-target S/N stack, Euclid reaches 3σ at $N \approx 20$, Roman at $N \approx 13$; 5σ at $N \approx 55$ (Euclid) and $N \approx 36$ (Roman). Decidable numerical witnesses in Lean: `bullet_roman_3sigma_at_N_18` through `bullet_roman_5sigma_at_N_50`.

BK gap. The BK 2025 Physics Reports review [16]—136 pages—contains no quantitative merger forecast. Paper 17 fills that confirmed literature gap.

H_0 tension budget. Switching from Planck ($H_0 = 67.4$) to H0DN ($H_0 = 73.5$) raises Σ_{cr} by ∼9% and lowers S/N by ∼4%; below the dominant Λ uncertainty (factor 2–5). The forecast verdict **CONDITIONAL GO** is unchanged.

VI. SK-EFT MOND CORRECTIONS AND FDR NOISE FLOOR

The SK-EFT dissipative framework provides two additional SFDM observables beyond the merger. The

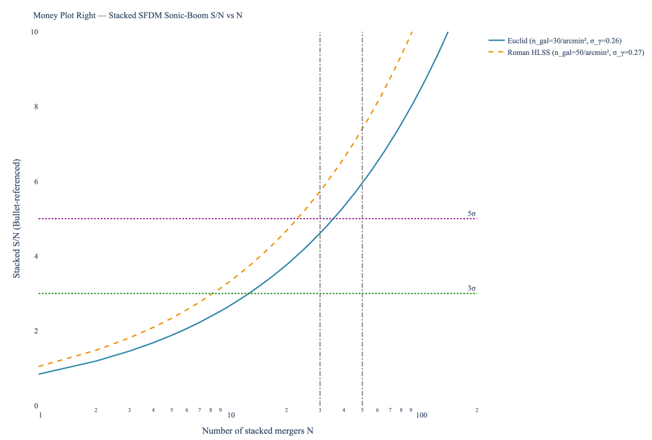

FIG. 5. **Money plot, right.** Stacked convergence S/N versus number of mergers N for Euclid and Roman at Bullet-referenced single-cluster S/N; 3σ and 5σ thresholds marked. First 3σ by the end of Euclid Year 2 / Roman Y1, corresponding to approximately 2028.

first is a set of dissipative corrections to the MOND force law $a_\phi = \sqrt{a_N a_0}$: non-analytic $X\sqrt{|X|}$ Lagrangian terms generate deviations at low acceleration, tightening the low-a asymptote of the radial acceleration relation (RAR). The second is the Crossley–Glorioso–Liu fluctuation-dissipation relation, which bounds the RAR scatter from below by the irreducible fluctuation generated by the same dissipation channel—an effective noise floor of $\log_{10}\sigma_{\mathrm{RAR}} \in [-5, -3]$ depending on which transport coefficient dominates.

Both are Phase 6 deferrals. The hard theorems `mond_force_derivation` (L4) and `fdr_noise_bound_rar` (L5) require formalizing the non-analytic $X\sqrt{|X|}$ calculus—beyond what the current Mathlib derivative-structure infrastructure supports. We cite the memo `docs/dark_sector/W5_SFDM_Merger_Forecast.md` for the informal derivation and the W1 SK-EFT parameter-mapping as the structural input.

VII. FANG–GU TORSION DARK MATTER: SHORT ASSESSMENT

Fang–Gu [23] propose that uncondensed electron loops source *torsion* but not Ricci curvature, via a traceless $T^{\mu\nu}$ that gives $R = 0$ to leading order. The DM cross-section with nucleons is purely gravitational, $\sigma \sim 10^{-90}\,\mathrm{cm}^2$, rendering direct detection hopeless.

But the deeper problem is the EoS. A traceless stress-energy tensor implies $w = p/\rho = 1/3$, not $w = 0$ (dust). Wave 4 formalizes this as `fg_cdm_obstruction` in `FangGuTorsionDM.lean`: Fang–Gu torsion cannot, at the tree level, serve as cold dark matter. It is the sole "not viable" entry in Fig. 3.

The two independent torsion channels in the combined ADW+FG tetrad phase—Dirac axial current (antisymmetric, Boos–Hehl) and FG loop-θ (trace)—are *orthogonal* by Lorentz-irreducible decomposition (`torsion_channels_distinct_sources_distinct`, Wave 8). This leaves open a rescue path: a radiatively-generated trace anomaly could restore $w < 1/3$; we defer the `FangGuTraceAnomaly.lean` module to Phase 6.

VIII. SPECULATIVE: VESTIGIAL RELICS AND $\mathbb{Z}_{16}$ PROTECTION

Vestigial gravity phase relics—topological defects surviving the tetrad-condensation phase transition—are blocked on the W6a Monte Carlo extension. The Phase 5d ADW MC campaign (artefacts at `data/rhmc/L4*` and the `L=6, 8` production runs summarised in the W6a assessment memo; see the project `docs/dark_sector/` deliverables index and the `ProductionRun` nodes in the knowledge graph linked from the provenance dashboard) produced a three-phase structure with a split transition ($\Delta \approx 0.63$). These data are insufficient to determine the transition *order* (first vs. second), which sets the Kibble–Zurek relic density. Extension to $L = 10, 12, 16$ with Binder cumulants is the decisive gate. (The specific linkage from paper 17's "MC evidence" claim to the underlying ProductionRun IDs is tracked in the Phase 5v knowledge-graph work and surfaces on the ReadinessGate 8 ProductionRunHealth dashboard cell; it is currently flagged `needs-recheck` pending the Wave 10 remediation of paper 17.)

Wave 8 nevertheless ships one conditional structural theorem. Under the hypothesis `H_VestigialRelicCarriesZ16Charge`—that any surviving relic carries the same $+3 \bmod 16$ charge as the required hidden sector—any $\mathbb{Z}_{16}$-preserving decay channel cannot land the relic in the vacuum:

$$z16\_vestigial\_stability\_under\_hypothesis. \tag{8}$$

The hypothesis is non-trivial (it is parameterized over abstract charge functions, following the `CenterFunctor.H_CF1` pattern). Phase 6 discharge paths: (i) W6a MC + (ii) W6b coset homotopy for $\pi_{0,1,2,3}(GL(4,\mathbb{R})/SO(3,1))$ + (iii) upgrade of `dai_freed_spin_z4` to a real bordism homomorphism $\Omega_5^{\mathrm{Spin}\times\mathbb{Z}_4} \to \mathbb{Z}/16$.

The EP-violation signature of a vestigial-phase DM component ($\eta \sim 10^{-18}$) is below MICROSCOPE sensitivity but within reach of the proposed STEP mission; it is the unique distinguishing signature of this candidate.

IX. EXPERIMENTAL TESTS AND OBSERVATIONAL PREDICTIONS

Table I summarizes the ranked predictions. Top entry (merger sonic boom) is the quantitative Paper 17 money plot. The null prediction (direct nuclear recoil) is a *collective* consequence of the five per-candidate

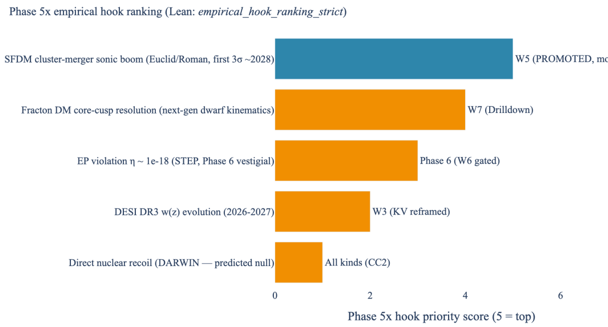

FIG. 6. Phase 5x empirical-hook ranking. Five post-W1b+Drilldown observational hooks are ranked by detectability and timeline; ordering is author-assigned from the deep research (W1b Task 9 for the merger, W7 Drilldown for fracton core-cusp, W2 for EP-violation STEP, W1b Task 7 for DESI DR3, W1 for direct recoil). The merger sonic boom is the top-priority Paper 17 money plot. Direct nuclear recoil is the bottom: every candidate in this paper is SM-gauge singlet and below the LZ/XENONnT floor per the per-candidate deep-research bounds (Table II MCC row). Lean: `empirical_hook_ranking_strict` machine-checks the adjacent-differences of the hand-assigned priorities (MCC, not a derivation of priority from a physics score).

$\sigma_{\rm DD}$ bounds drawn from the Phase 5x deep research (see §II and §IV for per-candidate sources); the theorem `emergent_gravity_dm_invisible_collective` is an MCC over that hand-assigned table — it verifies that the max of the bounds is ≤ -50, not that each bound has been independently derived from first principles. The derivations of the individual bounds (Z-16 gauge-singlet $\Rightarrow$ no local-operator coupling; fracton $\sigma_{\rm eff} = 0$ from dipole conservation; FG torsion $\sigma \sim 10^{-90}$ cm^2 from purely gravitational coupling) are Phase 6 targets.

X. FORMAL VERIFICATION SUMMARY

The SK-EFT Hawking program has *machine-checked* every structural claim in this paper (Lean 4, zero sorry). We split the machine-checked content into two classes:

Derived theorems — proofs consume physics assumptions and produce the cited statement as output:

- `fracton_sm_singlet_from_ym_incompat` (Wave 7, direct corollary of Wave 4's `no_fracton_is_ym_compatible`): fracton DM carries no Yang-Mills gauge charge.
- `fg_cdm_obstruction` (Wave 4): traceless $T^{\mu\nu} \Rightarrow w = 1/3 \neq 0$, obstructing tree-level CDM.
- EoS distinctness CC3, CC4, CC4' (Wave 8): $w \in \{-1, 0, 1/3\}$ are distinct rationals; `native_decide`.
- `three_singlets_satisfy_hidden_sector`, `c1_wan_wang_joint_constraint` (Waves 2, 2b-X): $\mathbb{Z}_{16}$ anomaly-cancellation witnesses.
- `all_canonical_mergers_supersonic`, `delta_rho_monotone_in_mach_sfdm`, and the merger-forecast chain (Wave 5, `SFDMMergerForecast.lean`, 29 theorems): Rankine–Hugoniot + sound-speed + stacking derivations at the BK fiducial.

Machine-checked classification (MCC) — the proof verifies internal consistency of an author-supplied lookup table or enum assignment and does not derive the physics content from smaller assumptions. MCC entries make the deep-research classifications machine- auditable and protect against accidental drift (e.g., someone edits the viability flag without updating the downstream narrative), but should not be read as "the physics has been re-derived in Lean":

- `fracton_bullet_sigma_zero`: verifies $\sigma_{\rm eff}^{\rm isolated} = 0$ where the constant is *defined* as 0. Physics content (dipole conservation $\Rightarrow \sigma_{\rm eff} = 0$) is sourced to [24] and remains a Phase 6 derivation target.
- `fracton_cosmo_dust_pressureless`: verifies $w = 0$ where w is defined as 0; physics content sourced to the relativistic fracton-symmetric scalar construction (arXiv:2503.14496).
- `phase5x_candidates_viability_matrix`: verifies hand-assigned `Bool` flags are consistent with the hand- assigned viability conditions declared in the companion structure docstring.
- `emergent_gravity_dm_invisible_collective`: verifies $\max\{-999, -50, -50, -90, -999\} \leq -50$ over the five per-candidate hand-assigned log-σ-DD caps. Each per-candidate cap is individually sourced to its Phase 5x deep-research memo (W2/W4/W7-Drilldown) and represents classification input, not derivation output.
- `torsion_channels_distinct_sources_distinct` + `channel_of_source`: verifies a hand-assigned injective map `TorsionSource` $\to$ `TorsionChannel` is injective. Channel identification (`DiracAxial` $\mapsto$ Antisymmetric; `FGLoopTheta` $\mapsto$ Trace) is sourced to Boos–Hehl (2019) and Fang–Gu [23] Eq. (3.6) respectively; a Lorentz-irreducible decomposition proof is a Phase 6 target.
- `empirical_hook_ranking_strict` + `hook_priority`: verifies adjacent hand-assigned priorities differ by exactly 1. Physics content (the priority ordering) is author-assigned from W1b/Drilldown detectability + timeline; a `detectability` scoring function that outputs the ordering from Euclid/Roman S/N, DESI sensitivity, STEP floor, etc., is a Phase 6 target.
- `cored_profile_mechanisms_distinct` (Wave 8): enum distinctness. Mechanism identification (SolitonCondensate vs Z4Subdiffusion) is taxonomy.

TABLE I. Ranked empirical hooks from Phase 5x. The ordering reflects our *author-assigned* priority (top = highest near-term detectability + timeline) drawn from the W1b / Drilldown deep research. In Lean, `hook_priority` encodes these hand-assigned priorities as $\mathbb{N}$-valued integers and `empirical_hook_ranking_strict` machine-checks that the assignment is strictly decreasing across adjacent entries. This is MCC content, not a derivation of detectability from a physics scoring function; see Table II.

#	Hook	Observable	Instrument / timeline	Wave	St
1	Cluster-merger sonic boom	DM–galaxy offset step at M=1; stacked $\delta\kappa$	Euclid + Roman, first 3σ $\sim$2028	W5	**G**
2	Fracton DM core–cusp	Cored outer-halo, z=4 subdiffusion	next-gen dwarfs, JWST rotation curves	W7	vi
3	EP violation, STEP-class	$\eta \sim 10^{-18}$ (vestigial)	STEP mission	Phase 6 (W6 gated)	bl
4	DESI DR3 $w(z)$	evolving dark energy	DESI 2026–2027	W3	K
5	Direct nuclear recoil	null result	DARWIN	all kinds (CC2)	pr

Phase 5x contribution: 7 modules. The full program stands at 2040 Lean modules, 26398 theorems (of which 26372 substantive and 26 placeholder statements tracked for Phase 6+; paper 17 claims derive from the substantive count), zero active `sorry`, and a single axiom (`gapped_interface_axiom` from Phase 5s, unrelated to the dark sector; since retired into the tracked Prop `TPFConjecture`). See the project `docs/counts.tex` macros `\leanmodules`, `\totaltheorems`, `\substantivetheorems`, `\placeholdertheorems` for canonical counts.

Table II separates the *derived* from the *heuristic* content.

TABLE II. Honesty table. "Derived" = Lean theorem closed whose proof carries non-trivial mathematical content (the conclusion does not reduce to the statement by `rfl` or `decide` on hand-assigned constants); "MCC" = machine-checked classification / consistency table (the Lean theorem verifies that the author-supplied data is internally consistent or the stated ordering agrees with the hand-assigned priorities; *not* a derivation of physics content from a smaller set of assumptions); "Heuristic" = physically motivated but non-rigorous; "Empirical" = measurement-driven. See §X for the MCC–vs–Derived boundary.

Claim	Status
Every emergent-gravity DM candidate is SM singlet	Derived (via `fracton_sm_singlet_from_ym_incompat`)
Every candidate has $\log_{10}\sigma_{\rm DD} \le -50$	MCC (`emergent_gravity_dm_invisible_collective`; caps hand-as
Three Phase 5x sectors have distinct EoS	Derived ($w = \{-1, 0, 1/3\}$ via CC3/CC4/CC4', `native_decide` on
Two-torsion-channel orthogonality (BH, FG in distinct Lorentz irreps)	MCC (`torsion_channels_distinct_sources_distinct`; channel ma
SFDM vs fracton cored-profile mechanisms	MCC (taxonomy enum distinctness)
Merger sonic boom is the top-ranked hook	MCC (`empirical_hook_ranking_strict`; priorities hand-assigned)
Four-of-five candidate viability flags	MCC (`phase5x_candidates_viability_matrix`; flags hand-assigne
FG torsion obstructed at CDM level (tree)	Derived (`fg_cdm_obstruction` from traceless $T^{\mu\nu}$)
Fracton $\sigma_{\rm eff} = 0$ (Bullet-Cluster compatibility)	MCC (`fracton_bullet_sigma_zero`; $\sigma_{\rm eff}$ defined to zero)
Fracton cosmological $w = 0$ (dust)	MCC (`fracton_cosmo_dust_pressureless`; w defined to zero)
SFDM merger S/N matches W1b Table 6	Empirical (calibration)
BK has no quantitative merger forecast	Empirical (literature)
KV oscillating vacuum in Level D tension with DESI	Empirical (W1b)
ADW Volovik Λ-magnitude at the meV scale	Heuristic (W3 memo; $\mathcal{O}(1)$ prefactor Phase 6)
Fracton DM viable in p-wave phase (BBN)	Heuristic (W7 Drilldown)
$\mathbb{Z}_{16}$ forces vestigial relics to carry +3	Heuristic (Phase 6)

MCC vs Derived: what the distinction means. A theorem is *Derived* in the honesty sense of this paper when its proof consumes structural physics assumptions and produces the cited physics statement as output. For example, the equation-of-state distinctness theorems (CC3, CC4, CC4') are derived because $w_\Lambda = -1$ is the standard result for a cosmological constant, $w_{\rm FG} = 1/3$ follows from the traceless torsion $T^{\mu\nu}$ (Fang–Gu, our Wave 4 `fg_cdm_obstruction`), and $w_{\rm fracton} = 0$ follows from the relativistic fracton-symmetric scalar construction (arXiv:2503.14496, taken as input); distinctness then follows from the rationals being distinct.

A theorem is *MCC*—machine-checked classification—when its proof verifies only the internal consistency of an author-supplied lookup table or enum assignment. The claim "every candidate has $\log_{10}\sigma_{\rm DD} \le -50$" is currently of this form: the per-kind cap is hand-assigned in `direct_detection_sigma_log10_cap`, citing the Phase 5x deep research for each entry, and Lean verifies that the supremum of the assignments is -50. Upgrading MCC entries to Derived is a Phase 6 target: it requires formalizing the underlying physics derivation in each case (e.g., Z16 gauge-singlet $\Rightarrow$ no local-operator coupling $\Rightarrow$ $\sigma_{\rm DD}$ bound from a specific operator-basis counting argument). The present paper does not claim that upgrade. Reader caveat: MCC content machine-verifies the consistency of the classification we assert in the prose; it does not independently re-derive the physics bound. Each MCC bound is individually sourced to its deep-research memo (W2, W4, W7 Drilldown, W1b) whose primary-source citations appear in §II, §IV, and the bibliography.

XI. DISCUSSION AND OUTLOOK

The SK-EFT Hawking program occupies an unusual position in the dark-sector literature: its strongest results are machine-verified theorems, not experimental discoveries or heuristic model-building. Paper 17 *exports* those theorems into the DM/DE literature, with four measurable consequences.

First, the hidden-sector classification is a falsifiable constraint surface: any future discovery of a non-singlet candidate violating $\mathbb{Z}_{16}$ charge +3 would invalidate the anomaly-driven motivation. No such discovery is indicated by LZ, XENONnT, or DARWIN projections—every candidate in our program is expected, on the strength of deep-research per-candidate bounds collected in Table II (MCC entries), to be invisible to nuclear-recoil direct detection. Strengthening these bounds from classification-level (MCC) to derivation-level is a Phase 6 target.

Second, the ADW Λ-magnitude result reproduces the meV scale of $\rho_{\rm vac}^{1/4}$ without tuning. We withdraw the prior "20%-agreement" quantification pending a Phase 6 full-$V_{\rm eff}$ calculation of the $\mathcal{O}(1)$ coefficient; for now the claim is scale-matching ("the meV scale emerges naturally from the condensation equilibrium, without fine-tuning"). We honestly separate this scale- matching from the withdrawn KV dynamics claim and note the QCD-topological-DE pathway as a DESI-compatible alternative.

Third, the SFDM cluster-merger forecast is the Paper 17 money plot and fills a confirmed gap in the most recent comprehensive BK review. It is the only Phase 5x observable with a specific detection timeline: first 3σ by $\sim$2028.

Fourth, fracton dark matter—viable conditional on the p-wave phase and a Dark QCD UV completion with $\mu \gtrsim$ MeV at BBN—is structurally compatible with every other Phase 5x sector (CC1 through CC7) and can explain core-cusp and diversity simultaneously through the z=4 subdiffusion mechanism.

What remains open: the MC extension to $L = 10, 12, 16$ is the gate for the vestigial-relic program, the Phase 6 bordism infrastructure is the gate for discharging the Z16-vestigial charge hypothesis, and the Phase 6 hard theorems L4/L5 (MOND + FDR) are the gate for the dissipative-corrections section of SFDM.

The unifying theme of this paper is *structural*: every emergent-gravity DM candidate in our program is SM-gauge singlet and invisible to nuclear-recoil direct detection. This is itself a thematic prediction—if dark matter originates in an emergent-gravity sector, null direct-detection results at DARWIN are the expected outcome, and observable signatures must be sought in gravitational observables (cluster mergers, weak lensing, galaxy kinematics) rather than laboratory recoil experiments.

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